7 **(a)** **(i)** The photoelectric effect refers to the emission of electrons from a metal surface when the surface is irradiated with electromagnetic radiation of a high enough frequency. B1

(ii) Allow 380 nm or 390 nm A1

(iii) Threshold frequency is determined when there is no responsivity from photocathode/ no release of photoelectrons. B1

From Fig. 7.2, threshold wavelength is approximately 640 mm. Threshold frequency is found by using speed of light divided by threshold wavelength (or 630 nm) B1

*Statement to be updated

(iv) Using hc/λ = work function + $E_{k,\ max}$, B1

$hc/(450 \times 10^{-9}) = h\ (4.76 \times 10^{14}) + 0.5(9.11 \times 10^{-31})v^2$ C1

$v = 5.27 \times 10^5\ m\ s^{-1}$* A1

(v) Using a low work function photocathode will allow electrons to be emitted easily with photons of larger wavelength. B1

(b) **(i)** Current = nq/t

$7.8 \times 10^{-4} = (n/t)*(1.6 \times 10^{-19})$ C1

$n/t = 4.88 \times 10^{15}\ s^{-1}$ A1

(ii) Number of photons per unit time

$= 4.88 \times 10^{15}\ (100)$ A1

$= 4.88 \times 10^{17}$

(iii) 4.8 mA of current is detected from photocathode when 1 W of power of EM radiation is incident on it. B1

(c) **(i)** Use wavelength = 225 nm, R = 9.0 mA/W and wavelength = 540 nm, R = 22 mA/W to correctly determine η at 5 %. B2

(ii) Correct computation of values with maximum values of 2sf A2

To deduct one mark for one wrong calculation.

***η* / %**	***λ* / nm**	***R* / mA W^{-1}**
3	300	7.2
3	400	9.7
3	500	12
3	600	14
3	700	17

(iii) Correct plotted points with best-fit curve B2

(d) **(i)** If there is little magnetic shield, magnetic field will curve electron paths, steer the electrons away from the dynodes. B1

The voltage used in PMT is higher than 1000V and must be electrically insulated for safety reason.

Accept any reasonable answer.

(ii)
- Use in eye devices to measure interruption of light. B1
- To detect radiation.
- Measure intensity and spectrum of light emitting materials in laboratory.

https://en.wikipedia.org/wiki/Photomultiplier_tube#Usage_considerations